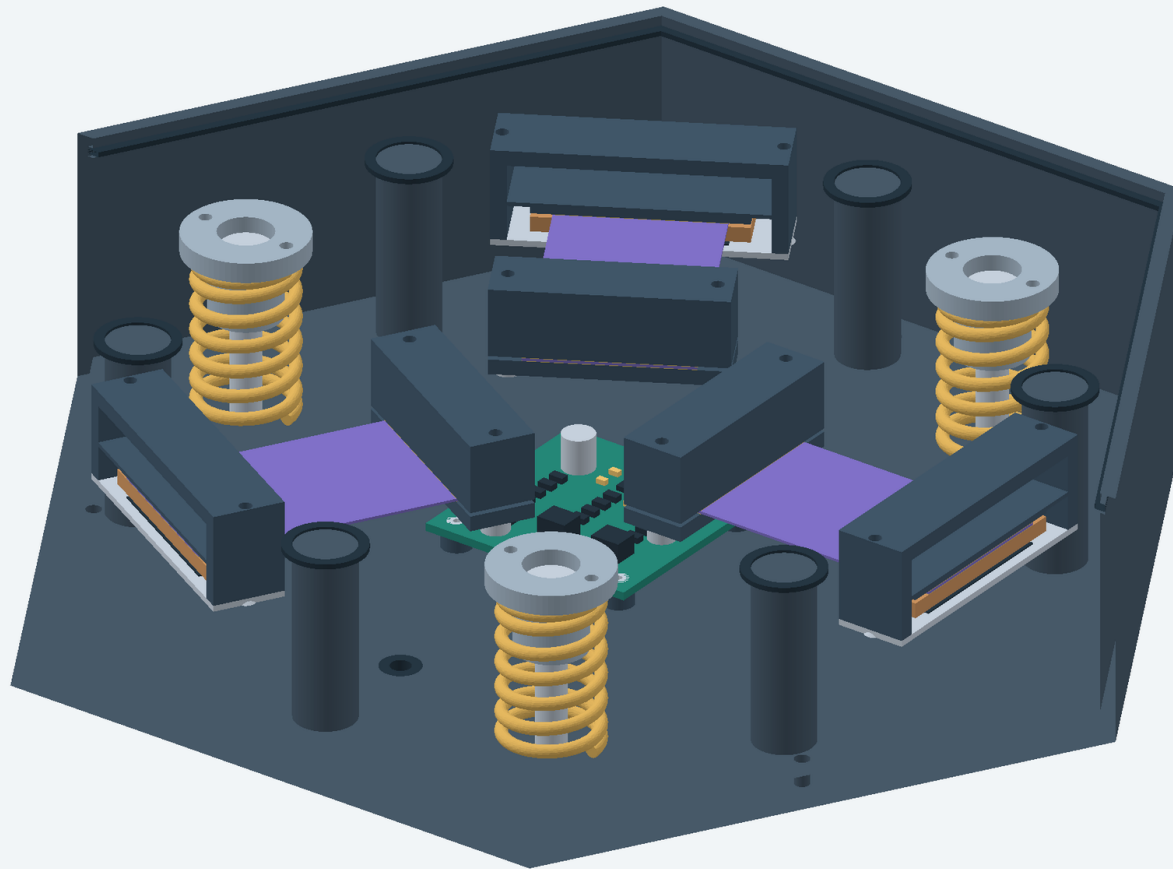


A complete tile, from tread to transducer

Regular hexagon • 200 mm across flats • 50 mm high • 2 mm cover stroke



Based on your reference

The cover carries three piezoelectric cantilevers with tip masses. During a footstep, acceleration of the cover makes each mass lag and bends its beam.

Two separate mechanical roles

Springs return the cover. Six structural stops transfer the main foot load to the pan after the cover moves 2 mm. The ceramic beams are not foot-load supports.

What is defined

142 CAD components: housing, cover, tread, guide assemblies, springs, beam cartridges, limiters, fasteners, seals and a reference electronics bay.

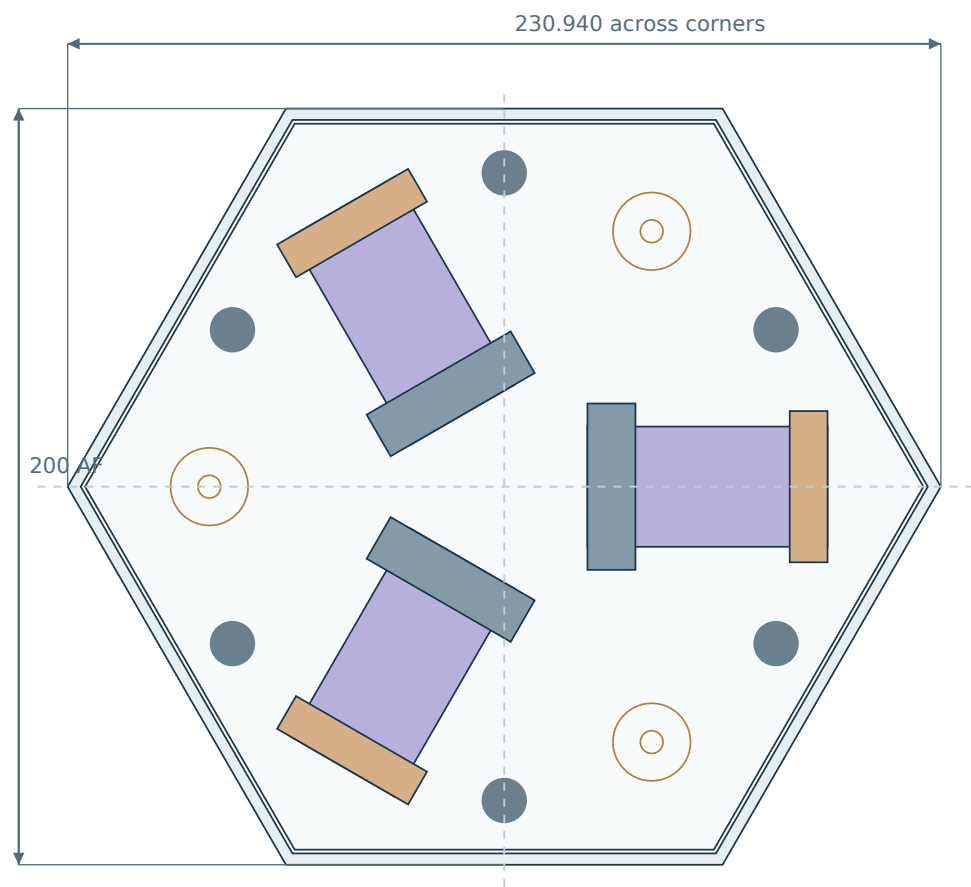
What accuracy means here

The STEP geometry and clearances are dimensioned. Spring specification, clamp details, PCB layout and load/energy performance still require prototype qualification.

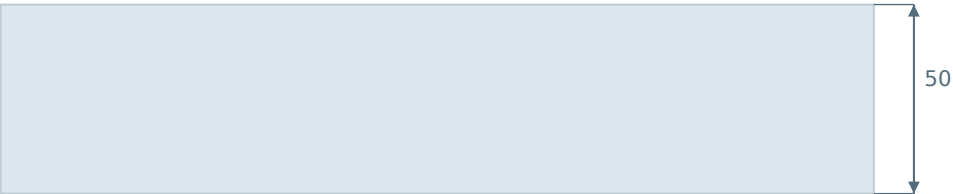
Use for the CSP and bench development. This is a detailed engineering concept, not a certified floor tile or a fabrication release. The user reference is a schematic, so unspecified dimensions were designed here rather than inferred as measurements.

Envelope and installation geometry

Plan and elevation at 1:2 when printed at 100% on A4 landscape; dimensions override display scaling.



PLAN - cover removed to show layout



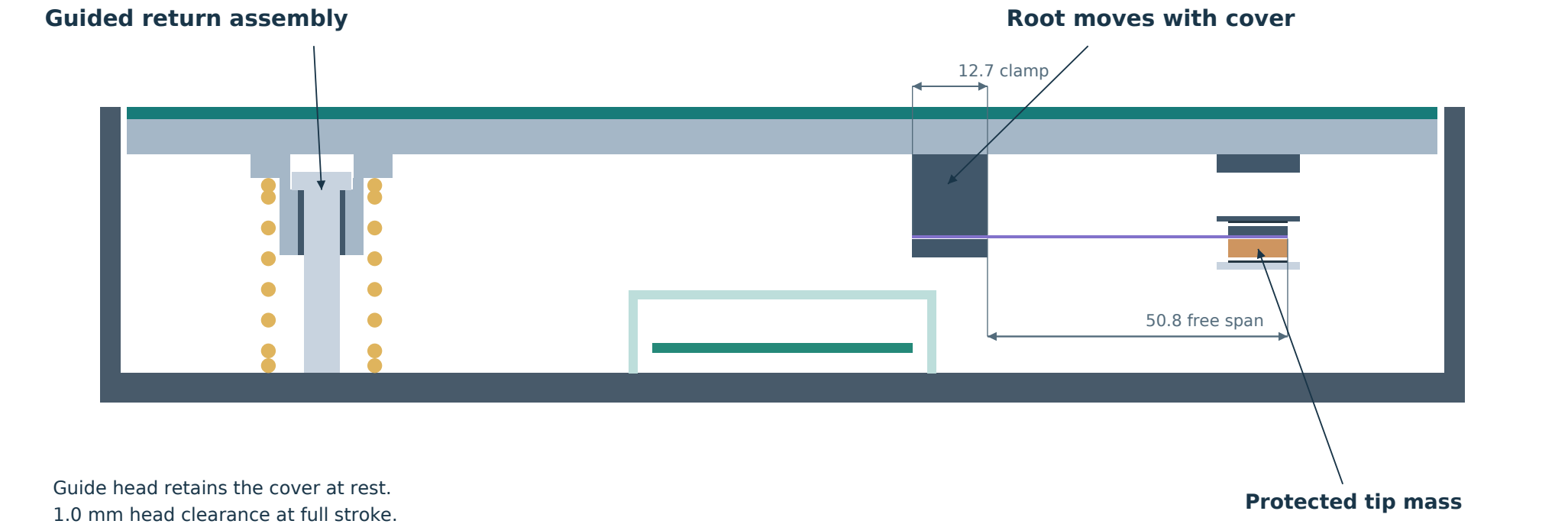
ELEVATION - fixed rim defines overall height

Design dimension	Value
Width across flats	200.000 mm
Width across corners	230.940 mm
Hexagon side length	115.470 mm
Housing wall / base	3.0 / 5.0 mm
Cover AF / thickness	192.0 / 6.0 mm
Tread thickness	2.0 mm
Cover-to-wall clearance	1.0 mm at each flat
Guide stations	R78; 60°, 180°, 300°
Beam axes	0°, 120°, 240°
Structural stop stations	R83; 30° + n × 60°

Width assumption: “20 cm width” is interpreted as across opposite flats. A 200 mm corner-to-corner hexagon would be only 173.205 mm across flats and needs a redesigned internal layout.

Section: the cover moves, the base stays fixed

Nominal undeformed geometry. Diagram shows the beam axis and opposite guide; strokes are not exaggerated.

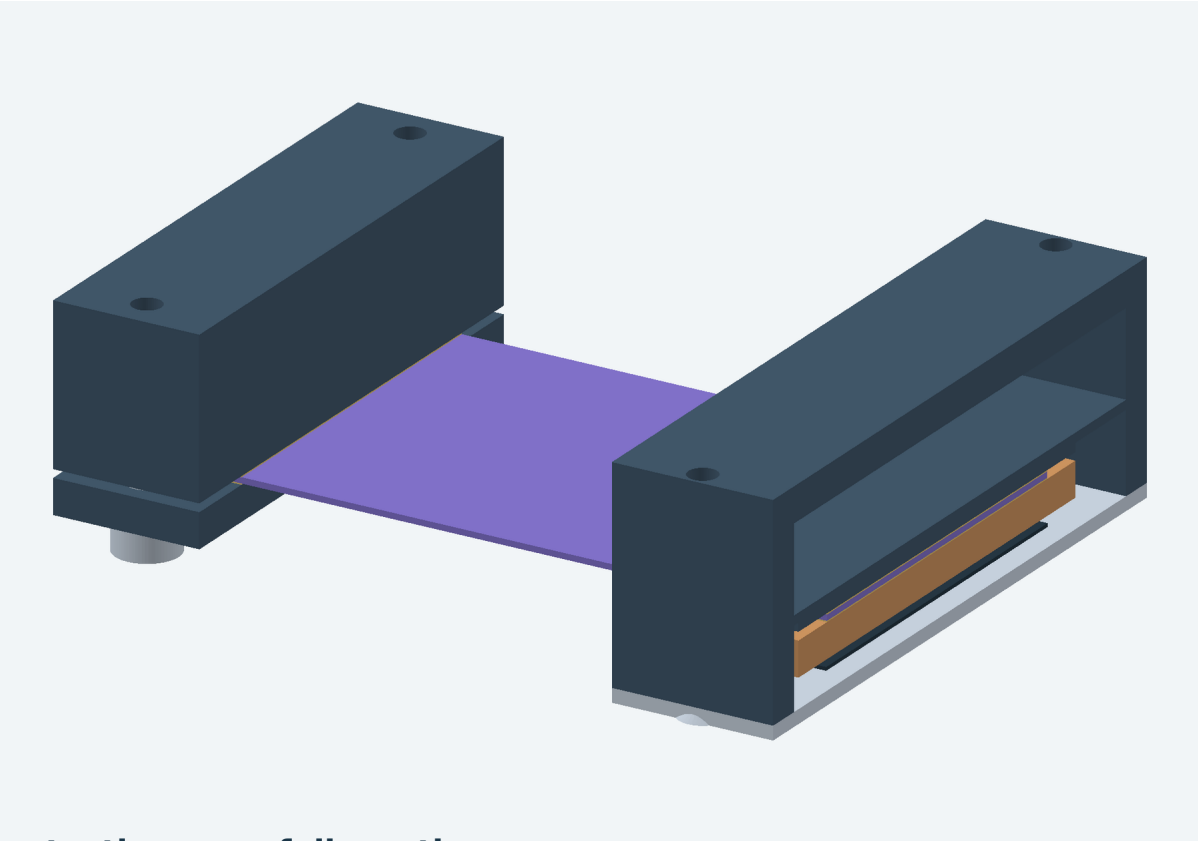


Vertical datum	z at rest / mm
Underside of tile	0.000
Pan floor surface	5.000
Piezo mid-plane	28.000
Spring upper seat	38.000
Metal stop top	40.000

Vertical datum	z at rest / mm
Bumper top	40.500
Moving plate underside	42.000
Plate upper surface	48.000
Tread / fixed rim upper surface	50.000
Plate underside after 2 mm stroke	40.000

Cantilever cartridge: three identical modules

Each cartridge is attached beneath the moving cover. All tip fasteners straddle the ceramic.



A protective cage follows the cover

The upper and lower pads travel with the beam root. They limit relative mass motion; they do not define the 2 mm cover stroke. Screws are outside the beam edges, so no holes are drilled through PZT.

Tip rotation matters. With a 45.8 mm lever arm, a 0.45 mm centre displacement gives about 0.845° pitch. The minimum nominal edge clearance is about 0.0765 mm. This is a geometric check, not an impact or fatigue guarantee.

Feature	Nominal
Beam envelope [1]	63.5 × 31.8 × 0.51
Clamped length	12.7
Free beam length	50.8
Mass-centre lever arm	45.8
Proof-mass blank	10 × 40 × 3
Tip cap	10 × 40 × 1.5
Electrical shims	0.10 each side
Upper / lower flat gaps	0.60 / 0.60
Visual tip-centre range	±0.45

Source and modelling boundary
The selected reference bender is Piezo.com T220-A4BR-2513XB, a PZT 5A/brass bimorph [1]. It is modelled as its published external envelope. Internal adhesive/electrode thicknesses are not reverse-engineered. Clamp preload and permissible tip mass need supplier confirmation.

Spring and beam calculations

Screening estimates for understanding and prototype selection; these are not measured output or strength results.

Cover return springs

Three custom target springs: wire diameter $d = 2.5\text{ mm}$; mean coil diameter $D = 18\text{ mm}$; five active turns, seven total turns; assumed shear modulus $G = 79,000\text{ N/mm}^2$. Free length 34 mm , installed length 33 mm .

$$k = Gd^4 / (8D^3n) = 13.23\text{ N/mm per spring}$$
$$K = 3k = 39.69\text{ N/mm total}$$

Idealised quantity	Value
Initial spring force (1 mm preload)	39.69 N total
Spring force after 2 mm downstroke	119.06 N total
Additional force over the stroke	79.37 N
Minimum spring length at full stroke	31 mm
Approximate solid length, 7×2.5	17.5 mm
Increase in stored spring energy	0.159 J

Spring force is not the net foot force: the moving assembly weight and seal friction also act. Most walking loads will bring this low-travel design onto its stops. Springs store and return mechanical energy; that stored energy is not an electrical-output prediction.

Mass-loaded piezoelectric beam

Use a first-mode model about the gravity-deflected equilibrium:
 $m \ddot{q} + c \dot{q} + kq + \text{electrical coupling} = -m \ddot{y}$
Here y is cover motion and q is mass motion relative to the cover.
Cover acceleration drives the beam.

For an illustrative estimate, assume the supplier’s quoted stiffness of 0.388 N/mm applies to a 50.8 mm free span [1]. Moving the force application to 45.8 mm gives $k \approx 0.388(50.8/45.8)^3 \approx 0.530\text{ N/mm}$. This mounting assumption must be checked.

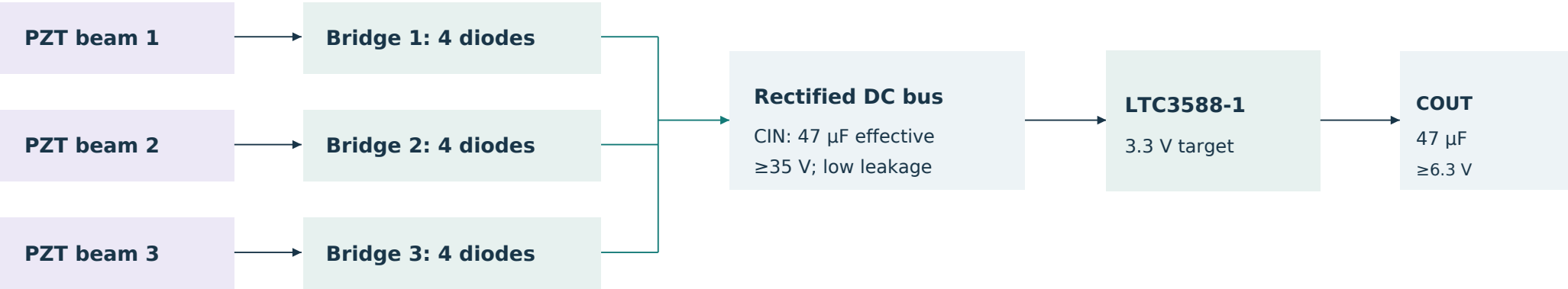
A 9.60 g steel mass blank plus cap/fasteners and effective beam mass gives roughly 12 g effective modal mass. Then $f \approx (1/2\pi)\sqrt{(k/m)} \approx 33\text{ Hz}$. Estimated static sag is around 0.2 mm ; CAD is shown flat before sag.

A walking cadence of a few steps per second does not continuously excite a 33 Hz resonance. The idea relies on the acceleration transient and subsequent vibration. Actual damping and the circuit strongly affect how much energy is collected.

Required measurement: use the measured capacitance and $\Delta E = \frac{1}{2}C(V_f^2 - V_i^2)$, then divide by counted activations. Subtract leakage and account for any load current.

Electrical architecture and connectivity

A complete reference topology with provisional board placement; not a routed or fabrication-ready PCB.



Why one rectifier per beam? The three beams need not vibrate in phase. Rectifying separately before joining the DC outputs avoids directly opposing their AC voltages. The LTC3588-1 internal bridge is bypassed here; the external DC bus feeds VIN. Verify this implementation and leakage on a bench [2].

Connection	Reference design
Each bridge, AC terminals A/B	D1: A → DC+; D2: B → DC+; D3: GND → A; D4: GND → B. Arrows denote anode → cathode.
Regulator input / unused AC inputs	VIN to rectified DC+; ground/exposed pad to GND. PZ1 and PZ2 left unconnected.
Buck and output	SW → 10 μH inductor → VOUT; COU between VOUT and GND; PGOOD available at test pad.
Rail capacitors / voltage selection	1 μF CAP-to-VIN; 4.7 μF VIN2-to-GND. D1 tied to VIN2, D0 to GND for 3.3 V.
Qualification points	Measure diode leakage, capacitor effective C and cold start. Allow for nominal ~5.05 V start-up at 3.3 V setting [2].

No battery is fitted. A low-duty-cycle sensor or switched LED is a candidate load only after energy measurement. The 44 × 44 mm board, parts and hood are mechanical placement envelopes; footprints, routing and exact capacitor packages remain provisional.

Materials and bill of materials

All 142 individual components are listed in BOM.csv and selectable in the 3D viewer.

Assembly / quantity	Material and function
1 housing with 6 integral stops	6061-T6 aluminium. 5 mm base, 3 mm sidewalls. Stops end at z40.
1 cover + 1 tread	6 mm aluminium load spreader; 2 mm replaceable EPDM tread.
3 guide sets	Steel headed guide, aluminium spring-seat cup, POM sleeve.
3 return springs	Custom spring-steel targets; 2.5 mm wire / 18 mm mean diameter.
3 piezoelectric beams	Reference PZT 5A/brass bimorphs; supplier external dimensions [1].
3 root clamp sets	POM tower + clamp shoe; polyimide electrical isolation.
3 proof-mass sets	Stainless steel mass + POM cap + isolating shims.
3 limiter cages	POM cage, removable steel bottom, two EPDM stop pads.

Material choices for the library corridor

The proposed rubber tread, polymer guide liners and buffered stops address contact noise, but their acoustic benefit has not been measured. A wipeable lip and drained pan address incidental water; the hood protects electronics only if its bond and feedthroughs are qualified.

Assembly / quantity	Material and function
6 stop bumper rings	EPDM; nominal compression 0.5 mm at full stroke.
1 wiper + 1 cable grommet	EPDM lip in the sidewall groove; underside cable port.
1 PCB + splash hood	FR4 reference board; polycarbonate hood bonded to pan.
18 cover screws	M3 countersunk reference geometry.
3 guide anchor screws	M4 countersunk nominal custom-length envelopes.
6 root / 6 tip screws	M3 root clamp / M2 tip clamp screws.
6 cage + 4 PCB screws	M2 nominal hardware envelopes.
Wires and electrical components	Fine flexible leads, three bridge rectifiers, regulator, capacitors, inductor and connectors.

Bio-based polymer samples can be investigated for removable cosmetic parts. This revision uses aluminium for the structural cover and stops because no creep, impact or moisture data have been supplied for a bioplastic formulation. Printed PLA copies are display models, not step-on products.

Assembly sequence and fit control

Prototype workflow. Dimensions are nominal; fit tolerances below are proposed targets requiring manufacturing review.

- 1

Prepare the pan

Machine the pocket, six stop pillars, wiper groove, drain holes and anchor bores. Deburr all edges; verify the six stop tops are coplanar.
- 2

Build the three cartridges

Fit insulating shims, root clamps and tip clamps. Keep fasteners outside the ceramic. Fit the limiter cages; confirm the top/bottom gaps optically.
- 3

Install captive guides

Insert each headed guide through its cup and sleeve. Seat springs and guides on the pan. Hold each guide in a fixture while securing its underside anchor screw.
- 4

Fit electronics and wiring

Assemble and bench-test the circuit separately. Install the board/hood and route flexible service loops clear of springs, beam spans and cage stops.
- 5

Close the tile

Lower the cartridge-loaded cover onto the three guide-cup flanges and secure its top screws. Verify return and retention. Fit the wiper and replaceable tread.
- 6

Qualify on a fixture

Use controlled mechanical loading on the bench. Check return motion, edge loading, electrical output and repeated cycling before considering foot use.

Fit / feature	Proposed target
Aluminium noncritical dimensions	±0.10 mm
Polymer noncritical dimensions	±0.15 mm; account for temperature/moisture
Stop top height / coplanarity	40.00 ±0.05 mm / within 0.05 mm
Guide shaft / sleeve bore	Ø6.00 −0.01/0; bore Ø6.10 +0.02/0
Radial guide clearance	0.050 to 0.065 mm, before alignment effects
Tip stop gaps	0.60 ±0.05 mm; measure assembled
Cover clearance at flats	1.0 mm nominal each side
Thread geometry	Tap sizes / thread classes set at manufacture; STEP threads are simplified

Assembly access matters. Lower guide-anchor screws are accessed from the underside. Cage bottoms are removable for beam service. Cover screws are accessed after peeling or removing the tread. The floor installation must allow the tile to be removed.

Supplier clamp instructions [3] are useful background, but their torque values do not automatically apply to these custom POM clamps. Establish preload experimentally with the actual beam and shims.

What has been checked, and what comes next

The CAD is inspectable and editable. Precision of geometry must not be mistaken for proven floor performance.

Check	Result / boundary
Solid validity	142 components valid in OpenCascade.
Nominal rigid interference	No unintended overlaps found at rest or at 2 mm cover stroke.
Envelope	200.000 mm flats; 230.940 mm corners; 50.000 mm height.
Tip pitch at ± 0.45 mm	Nominal minimum edge gap ≈ 0.0765 mm.
Excluded from rigid verdict	Thread engagement envelopes, PCB packages and indicative wire routing.
Not yet performed	FEA, local heel/edge loads, fatigue, wear, ingress/slip/acoustic tests or electrical-output tests.

Before any real corridor installation

Verify local cover bending, off-centre loading and guide binding; qualify the beam clamps, tip-mass retention and overtravel impacts. Check wet grip, raised-edge hazards, liquid paths and a removable flush mounting frame. No pedestrian load rating or IP rating is claimed.

Open the package

HTML: open HEX_200_Interactive.html in a current desktop browser; all viewer resources are embedded.
CAD: import CAD/HEX_200_Assembly.step in millimetres. Individual representative parts are included.
STL: units are mm; meshes are for visualisation/display prototypes.
Source: build_tile.py rebuilds the fixed Rev A design. Editing the width alone does not automatically redesign the mechanism.

Sources and provenance

[1] Piezo.com, T220-A4BR-2513XB

Published external dimensions, PZT 5A/brass construction and stiffness reference. Accessed 22 September 2026.

<https://piezo.com/products/piezoelectric-bending-transducer-t220-a4br-2513xb>

[2] Analog Devices, LTC3588-1 datasheet

Reference regulator connections and start-up behaviour. The proposed three-bridge circuit and board arrangement are this design’s engineering choices.

<https://www.analog.com/media/en/technical-documentation/data-sheets/35881fc.pdf>

[3] Piezo.com, Mounting Guidelines

Background on clamp practice and fixing security. Commercial-kit torque recommendations are not adopted for the custom clamps.

<https://support.piezo.com/article/126-mounting-guidelines>

[4] Piezo.com, Wiring Guidelines

Background on electrode connection and lead strain relief. Exact terminations need qualification with the purchased device.

<https://support.piezo.com/article/127-wiring-guidelines>

Original work: hexagonal geometry, spatial arrangement, spring target, captive guides, protective cages, clearances, diagrams and CAD were developed for this request. The supplied sketch inspired the mechanism; it did not provide a complete specification. Renders are derived from CAD, not AI-generated pictures.